Deletion, arrows, windows and composition: structural patterns in 10,404 certified polytope packings

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Abstract

The Polyhedra-in-Polyhedra catalogue holds 10,404 certified packings of n congruent convex polytopes in a scaled container, across 140 problem families in dimensions 3–6. The records are far from structureless. We sort their recurring patterns into eleven classes and group them by what can be said about them. *Mechanisms* are construction rules that we prove in Lean 4 for arbitrary bodies and containers: deletion and plateaus, subset relaxation ("arrows"), insertion, composition and recursion, lattice windows, and transfer by inclusion. *Descriptive patterns* are measured over the whole catalogue: layers and orientation classes, symmetry, algebraic scales and contact graphs. *Extremal patterns* are the 85 records proved optimal in Papers I and II of this series. We then measure how the classes relate. The formal implications are proved once: lattice windows give plateaus, composition with a single piece gives transfer bounds, and an extremal lower bound plus deletion proves a whole plateau optimal. Empirically, 92% of lattice windows lie on plateaus; about a quarter of the records with $n \leq 30$ contain a rattler, but lattice windows almost never do (lift 0.03); and no transfer bound beats a record in the 25 Platonic families, while 19 records are exactly a transfer. In dimensions 4–6, symmetric records are more than twice as common as in 3D, and depend strongly on the container: they are rare in simplices and common in cubes. The commonest rational relative rotations between orientation classes in 4D are the isoclinic $(60°, 60°)$ of the 24-cell's triality and simple 45° and 72° turns. PingYou Ltd's arrows reproduce their square-antiprism bounds from catalogue records, and applied to other antiprism families they improve 14 records, by up to 5%.

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1 Introduction

A *finite container packing* asks for the smallest scale $s(n)$ of a container C holding n congruent copies of a body K with disjoint interiors. The Polyhedra-in-Polyhedra catalogue [1] records, for 140 pairs (K, C) of Platonic, Archimedean and Catalan solids, prisms, antiprisms and regular polytopes in dimensions 4–6, the best certified packing found for each n up to 100: 8,514 records in 3D and 1,404, 279 and 207 in dimensions 4, 5 and 6. Every record carries an exact certificate of containment and pairwise separation. The packings come from soft-to-rigid continuation [5, 4], basin hopping, lattice cuts, composition, published constructions [6] and many community contributions, prominently those of PingYou Ltd, `konoha-ily2` and PolyPack.

Seen as data, the records are far from independent optima. They come in plateaus where $s(n)$ does not change. They are finite windows of lattice tilings, or compositions of smaller packings. They inherit large rigid blocks from neighbouring records. They show layered and symmetric arrangements, and their scales satisfy short algebraic equations. PingYou Ltd made two such observations explicit: an illustrated account of sixteen exact constructions and their number patterns [8], and "arrows" that recover smaller square-antiprism packings from a larger one [9]. This paper turns these observations into a classification with three layers:

1. **Mechanisms.** Each pattern class is a construction rule. Where the rule is a statement about packings, we prove it once in Lean 4 [11] for all bodies, containers and dimensions.
2. **Exact instances.** Records that realize a mechanism exactly are formalized at their exact scales, rather than at the clearance-padded scales of the certified records.
3. **Prevalence.** Each class is measured over the whole catalogue by reproducible scripts (`scripts/patterns/`), and every catalogue number in this paper is generated from their output.

1.1 Organization

Table 1 lists the eleven classes. The paper groups them in five parts:

- **Part I, Mechanisms** (Sections 2 to 7): the classes that are theorems about packings, P1–P5 and P11.
- **Part II, Descriptive patterns** (Sections 8 to 11): the classes that are measurements, P6–P8 and P10.
- **Part III, Extremal patterns** (Section 12): P9, the records proved optimal, summarized from Papers I [2] and II [3].
- **Part IV, Relations and dimensions** (Sections 13 to 16): how the classes relate, the patterns in dimensions 4–6, PingYou's sixteen constructions and a classification by family.
- **Part V, Formalization and limits** (Sections 17 and 18).

Each class section opens with a box giving its mechanism, its Lean statement and its prevalence.

1.2 Setting

Let E be a finite-dimensional real inner product space, $K, C \subseteq E$ and $s \geq 0$. A *packing* of n copies of K in sC is a family of affine isometries $g_1, \dots, g_n$ of E with $g_i(K) \subseteq sC$ and $\operatorname{int} g_i(K) \cap \operatorname{int} g_j(K) = \emptyset$ for $i \neq j$ (`Packing`). Reflections are allowed, so lower bounds proved for it hold for the catalogue's rotation-only packings. The catalogue measures s as the ratio of container to piece edge (or linear size for Catalan solids). A record r stores $s_{\text{full}}(r)$, which carries a clearance of at least 10^{-7} between pieces and walls, and an exact certificate. The touching limit s_{tight} is what the patterns below are about.

1.3 How to read the Lean links

Every formal statement is linked to its Lean declaration at commit `e3d934be83` of the repository [1]. All depend only on `propext`, `Classical.choice` and `Quot.sound` (Section 17).

Class	Mechanism	Formal statement (Lean)	Prevalence
Part I: mechanisms			
P1 Deletion, plateaus	$m \leq n$ pieces of an n-packing are an m-packing	`Packing.restrict`	861 plateaus, 3,397 records; 1,438 records built by deletion
P2 Arrows	subset, then F (fixed orientations, LP), then P (free); insertions, kicks	`Packing.restrict`, `Packing.of_separators`	PingYou's square-antiprism bounds reproduced; 14 antiprism records improved
P3 Insertion, completion	$(n-1)$-packing plus a piece in a hole	(constructive only)	1,974 records built by insertion
P4 Composition	$s_C(Q,n)\, s_Q(P,m)$ bounds $s_C(P,nm)$	`Packing.compose`, `Packing.pow`	15 exact composites, 63 nested records
P5 Lattice windows	Voronoi cells of distinct points	`voronoi_packing`, `lattice_window`	889 windows (407 at rational scale)
P11 Transfer	one piece of K in tK', or of C' in λC	`Packing.transfer_piece`, `Packing.transfer_container`	19 Platonic records equal a transfer, none beaten
Part II: descriptive patterns			
P6 Layers, orientation classes	a few layers, few orientations	(descriptive)	1,241 single-orientation 3D records
P7 Symmetry	invariance under container symmetries	(descriptive)	776 symmetric 3D records
P8 Algebraic scales	contact equations close the scale	(exact certificates)	1,624 closed forms
P10 Contacts	contact graph, rattlers	(measured)	649 of 2494 records ($n \leq 30$) with a rattler
Part III: extremal patterns			
P9 Extremal	lower bound meets construction	Papers I and II; `optimal_plateau_of_common_point`	85 records proven optimal

Table 1: The eleven pattern classes, in the order of this paper. Prevalence counts are over all 10,404 records, or over the 8,514 3D records where a geometric feature is needed.

Part I Mechanisms

2 Deletion, plateaus and frozen subsets (P1)

Mechanism. Any $m \le n$ pieces of an n-packing form an m-packing in the same container, so $s(m) \le s(n)$.
Lean. `Packing.restrict`, `Packing.mono`.
Prevalence. 861 plateaus with 3,397 records; 601 of 3,354 consecutive pairs are frozen subsets.

The simplest mechanism is deletion. The catalogue enforces its consequence, monotonicity: when a search finds $s(n+1) < s(n)$, the $(n+1)$-packing minus one piece replaces the n-record. 1,438 records arose this way, and 44 more from an explicit deletion move.

Plateaus. A plateau is a maximal run $n_0 < \cdots < n_1$ with $s(n_0) = \cdots = s(n_1)$. The catalogue has 861 plateaus containing 3,397 records. The longest are $s = 2$ for $2 \le n \le 32$ five-cubes (proved in Lean in Paper I [2]) and $2 \le n \le 24$ six-cubes (Hosono's computer-assisted centre lemma [7]). Others are $s = 4$ for $39 \le n \le 64$ cubes and $41 \le n \le 64$ triangular prisms (grid tilings), $s = 4$ for $37 \le n \le 56$ truncated cubes (the Cartesian shell, Section 6) and $s = 5$ for $75 \le n \le 93$ rhombic dodecahedra. A plateau ends at a *window number*, the largest count a rigid construction holds at that scale. Most plateaus are deletions of a single lattice window or tiling.

Frozen subsets. Deletion predicts a sharper relation between records. For records $k < N$ of one family, let $f(k, N)$ be the largest fraction of the k pieces that can be matched one-to-one with pieces of the N-packing after a single container symmetry and translation, with matching centres and identical placed vertex sets (orientation included, up to the piece's own symmetry; tolerance 0.02 edges; `scripts/patterns/inherit.py`). Then $f = 1$ means that record k is literally k pieces of record N, moved rigidly. Over 20,124 pairs with $N - k \le 6$ in 86 families, 1,384 pairs are frozen ($f = 1$), 601 of them consecutive ($N = k + 1$). Figure 1 shows the cube family. Frozen blocks appear exactly on the plateaus and on the k^3 grids. Between them, consecutive records share little rigidly: each was re-optimized after the deletion. Frozen pairs thin out with the gap: 601 of the 3,354 consecutive pairs are frozen, and fewer at each larger gap.

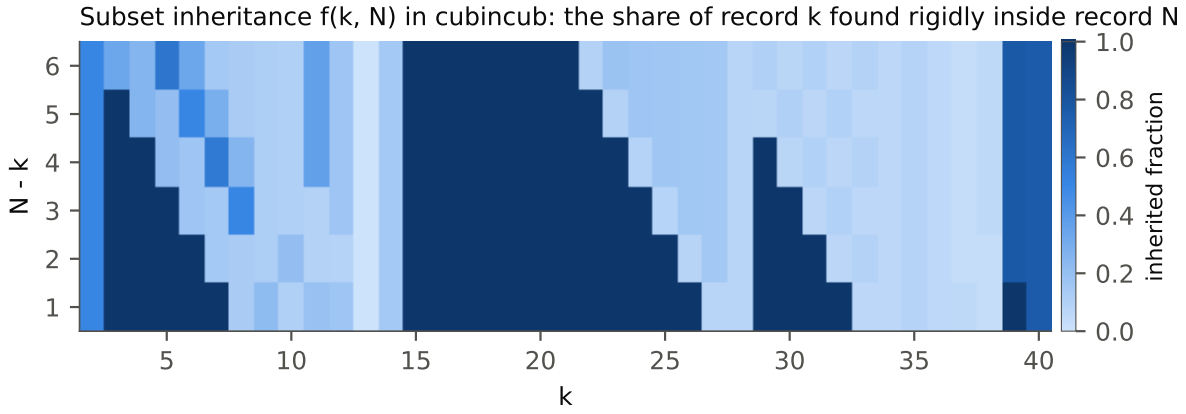


Figure 1: Subset inheritance in cubes in a cube. Dark cells: record k is a rigid subset of record N. The plateaus $s = 2$ ($n \le 8$), $s = 3$ ($15 \le n \le 27$) and $s = 4$ ($39 \le n \le 64$) appear as solid blocks.

3 Arrows: subset relaxation and kicks (P2)

Mechanism. Keep a subset of a larger packing (or add or turn a piece), then relax: F with orientations fixed, then P with everything free.
Lean. `Packing.restrict` for the subset, `Packing.of_separators` for the soundness of F.
Prevalence. PingYou's square-antiprism bounds reproduced from catalogue records; 14 records of other antiprism families improved.

PingYou Ltd observed that a larger packing contains good smaller ones that are *not* rigid subsets: selecting six or eight square antiprisms of their 19-piece packing and relaxing them recovers their 6- and 8-piece bounds [9]. They specified the relaxation as two steps, with the unit bodies fixed throughout:

F hold every orientation fixed and minimize the scale over the centres, re-selecting pair separation planes;

P minimize the scale with centres and rotations free, by a sequential linear program.

Arrows combine these with discrete moves: deletion (keep a subset), insertion (add a body at an affine guess or at the hub, the mean of the centres), and kicks (rotate one body about its own axis, or a group of bodies about an axis through their centroid, by $\pm 6^\circ$ to $\pm 16^\circ$), each followed by F then P.

3.1 F is an exact linear program

With orientations R_i fixed and one unit axis u_{ij} chosen for each pair, write $h_i(u) = \max_{x \in K} \langle u, R_i x \rangle$. Every constraint is then linear in the centres c_i and the scale s:

$$\langle N_f, c_i \rangle + h_i(N_f) \le \beta_f \, s \quad \text{(walls)}, \qquad \langle u_{ij}, c_i \rangle + h_i(u_{ij}) \le \langle u_{ij}, c_j \rangle - h_j(-u_{ij}) \quad \text{(pairs)}, \tag{1}$$

where $\{x : \langle N_f, x \rangle \le \beta_f\}$ are the container's facets.

Theorem 3.1 (Separator certificates; `Packing.of_separators`)**.** *If every placed piece lies in sC and every pair is split by a hyperplane, the placements form a packing.*

So any feasible point of (1) is a certified packing for that choice of axes. F iterates by re-selecting each pair's best separating axis at the current point; the point stays feasible, so the scale never increases (`scripts/patterns/fp.py`).

3.2 Reproduction from catalogue records

We ran arrows on the square-antiprism family ap4 using only catalogue records as sources. The searches tried subsets of records with $N \le 20$, chosen as caps along container directions, nearest clusters and random connected clusters, and then insertion and kicks from the improved packings (`scripts/patterns/arrows_search.py`, `arrows_kick.py`). Table 2 lists the results.

- For $n = 3$ and 5–9 the arrows reach PingYou's bounds to every digit they announced, starting from different packings: their 19-piece packing was not in the catalogue. For $n = 9$ their value is the closed form $1 + 2/(11 - 4\delta)$, $\delta = 1 + \sqrt{2}$.
- For $n = 6$ the subset alone stops at 2.2500. A single kick, a 16° spin of one body about its own axis followed by F and P, reaches their value, which they had obtained by selecting six bodies of their 19-piece packing.
- For $n = 10$ a single 6° spin of the record reproduces their tighter certificate. For $n = 11$–14 kicks improve the catalogue record but stay above PingYou's values. Those came from different constructions: relative square-face spins with receiver reflow, joint pair reinsertion, and ordered overlapping patches.

- The chain $6 \to 7 \to 8 \to 9$ follows their insertion arrows $N6 \to N7$ and $N8 \to N9$. When different starting points reach the same value, they have most likely reached the same rigid contact configuration by different arrows (Fig. 2).

PingYou Ltd has since submitted their constructions for $N = 2$–20 with exact certificates (pull request #107). They pass the catalogue's verifier and exact certifier and are now the records. They improve $N = 11$–20, with $N = 15$–17 at exactly 3, and give exact endpoint certificates for $N = 2$–10. For $N = 3$ and 5–9 their own witnesses replace our reproductions.

n	catalogue record	arrow from catalogue records	scale reached	PingYou Ltd [9]
3	1.973017314	subset $\{4, 6, 12\}$ of $N = 15$, F, P	1.9726490831	1.972649083082090
5	2.209265853	subset of $N = 9$, F, P	2.1785815015	2.178581501510501
6	2.292893610	subset of $N = 9$, F, P; spin 16°, F, P	2.2459217996	2.245921799598007
7	2.373563494	insert into $n = 6$ (affine guess); four spins, F, P after each	2.3629758095	2.362975810
8	2.402020630	insert into $n = 7$; four spins	2.3849425743	2.384942574336931
9	2.542195713	hub insertion into $n = 8$, F, P	2.4890416764	$1 + 2/(11 - 4\delta)$
10	2.590076603	spin 6° of the record, F, P	2.5900761895	2.590076189471171
11	2.743255848	kicks of the record, F, P after each	2.7406579330	2.652646434968480
12	2.807780046	kicks of the record	2.8008665059	2.764993777408391
13	2.907150931	kicks of the record	2.8905573692	2.842219125707920
14	2.970067187	kicks of the record	2.9543988095	2.945858498562152

Table 2: Arrows on square antiprisms in a square antiprism, from catalogue records only. "Scale reached" is the touching limit after F and P; the certified records add 10^{-7} clearance. The last column is PingYou's value. Their submitted constructions for $N = 2$–20 are now the catalogue records.

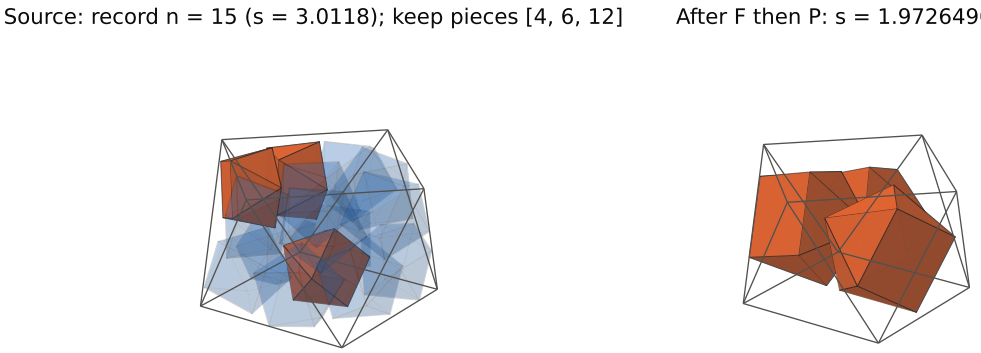


Figure 2: An arrow: three square antiprisms of the 15-piece record, then F and P, reach $S_3 = 1.9726490831$.

3.3 Arrows on other antiprisms

The same subset arrows, started only from catalogue records of the pentagonal, hexagonal, octagonal and dodecagonal antiprism families, improve 14 records, all certified. The largest gains are hexagonal antiprisms at $n = 6$, from 2.2466 to 2.1518, and octagonal antiprisms at $n = 9$ and 10, from 2.7464 to 2.6249 and from 2.8515 to 2.7118. Each came from a subset of a record with 11 to 13 pieces. Two of the new pentagonal-antiprism values match short closed forms, $(11 - \sqrt{5})/4$ at $n = 6$ and $(7 - \sqrt{5})/2$ at $n = 8$. These are conjectures until the contact systems are solved.

4 Insertion and completion (P3)

Mechanism. Add a piece to an $(n-1)$-packing, in its largest hole, and re-sharpen; or complete a rigid construction with pieces in its gaps.
Lean. Constructive only.
Prevalence. 1,974 records built by insertion.

The converse of deletion adds a piece to an $(n-1)$-packing and re-sharpens. A rigid version is *completion*: a construction with empty corners or gaps receives additional pieces at the same or slightly larger scale. PingYou's 61 truncated cubes [8] complete the 56-piece Cartesian shell (Section 6) with five corner pieces. The 40 wall and contact inequalities of their contact system sum to $5(2+\sqrt{2})(S-1) \geq 60$, so $S \geq 13 - 6\sqrt{2}$, and the construction attains it. That bound holds for the declared contact system, not for all packings. Insertion arrows (PingYou's $N6 \to N7$ and $N8 \to N9$, the latter with a hub body of orientation $\mathrm{diag}(1,-1,-1)$) are completion followed by F and P.

5 Composition and recursion (P4)

Mechanism. Scales are ratios, so they multiply: pack Q in C, then P in each copy of Q.
Lean. `Packing.compose`, `Packing.selfCompose`, `Packing.pow`, `triakis_recursion`.
Prevalence. 15 records equal a product $s(n)s(m)$ exactly; 63 records were built by nesting.

Theorem 5.1 (`Packing.compose`). *If n copies of Q pack in sC and m copies of P pack in tQ with $t \neq 0$, then nm copies of P pack in $(st)C$. In particular, for $K = P = Q = C$, $s(nm) \leq s(n)\,s(m)$ (`Packing.selfCompose`) and $s(n^k) \leq s(n)^k$ (`Packing.pow`).*

Proof. Conjugating g by the dilation $x \mapsto tx$ gives $\hat{g}\colon x \mapsto t\,g(0) + \vec{g}(x)$ with $\hat{g}(tA) = t\,g(A)$ (`dil`). The pieces $\hat{g}_i(h_j(P)) \subseteq t\,g_i(Q) \subseteq (st)C$ are disjoint within one g_i because $\hat{g}_i$ is an isometry. Across $i \neq i'$ they are disjoint because the interiors of $t\,g_i(Q)$ and $t\,g_{i'}(Q)$ are. □

Nested composition is how 63 records were built (`scripts/compose3d.py`, `compose_nd.py`; the whole-body inclusion idea is PingYou's contribution [10]). In the self-similar families, 15 records equal a product $s(n)s(m)$ exactly. They include the cube and prism tilings ($64 = 8 \cdot 8$ at scale 4) and the rhombic-dodecahedron record $36 = 6 \cdot 6$ at 4. The cleanest is the triakis tetrahedron, where PingYou recognized the 16- and 64-piece records as the 4-piece packing squared and cubed [8].

Theorem 5.2 (`triakis_recursion`). *For every $k \geq 0$, 4^k triakis tetrahedra pack in a triakis tetrahedron of scale $(5/3)^k$.*

Proof. Write the catalogue's triakis tetrahedron as $\{x : \langle m, x\rangle \leq 9\sqrt{2}/4\}$ over its twelve integer face normals m, of type $(1,3,1)$ (`Kt`). Place piece i at $R_iK + (\sqrt{2}/2)\sigma_i$ with $\sigma_i \in \{(-1,1,1),(-1,-1,-1),(1,1,-1),(1,-1,1)\}$ and rotations R_i with entries $\pm\frac{1}{3}, \pm\frac{2}{3}$. Containment in $\frac{5}{3}K$ is a linear consequence of the face inequalities. The six pairs touch along the planes $x_a \pm x_b = 0$. This gives the 4-piece packing at $5/3$ (`ktt4`), and Theorem 5.1 gives the powers (`ktt16`, `ktt64`). □

6 Lattice windows and tilings (P5)

Mechanism. Pieces that lie in the Voronoi cells of distinct points are disjoint, so every finite window of a lattice packing is a packing.
Lean. `voronoi_disjoint`, `voronoi_packing`, `lattice_window`; exact windows in `Windows.lean`.
Prevalence. 889 3D records are lattice windows, 685 with a single orientation and 407 at a rational scale.

Theorem 6.1 (`voronoi_disjoint`, `voronoi_packing`)**.** *If A lies in the Voronoi half-space of p against q and B in that of q against p ($p \neq q$), then A and B have disjoint interiors. Hence copies $g_i(K)$ lying in the Voronoi cells of distinct centres c_i, inside sC, form a packing. For a lattice Λ whose Voronoi cell contains K, any finite set of translates of K by points of Λ that fit in sC is a packing (`lattice_window`).*

The proof separates A and B by the bisector $\langle q-p, x\rangle = (\|q\|^2 - \|p\|^2)/2$. The theorem turns every lattice window into a finite containment check, and it covers multi-orientation windows whose pieces lie in the Voronoi cells of their centres. Two families are formalized completely (`Windows.lean`):

- **Rhombic dodecahedra.** $R = \{|x_i| + |x_j| \le 1\}$ is the Voronoi cell of D_3, the FCC lattice. For $x \in R$ and $v \in D_3$ one has $2\langle x, v\rangle \le \|v\|^2$: the support function of R is $\max(\|v\|_\infty, \|v\|_1/2)$, and both terms are at most $\|v\|^2/2$ on $D_3 \setminus 0$ (`rd_voronoi`). Every window $\{w \in D_3 : |2w_i + e_i| + |2w_j + e_j| \le M\}$ packs in $(M/2+1)R$ (`rdo_window`). The catalogue's records of 19, 44 and 93 pieces at scales 3, 4 and 5 are such windows: the FCC shell $1+12+6$ and the palindromic window $2+8+12+12+8+2$ of PingYou [8], and the 93-piece record. They are proved at the exact scales (`rdo19`, `rdo44`, `rdo93`), and by deletion every smaller count at the same scale (`lattice_windows`).
- **Truncated cubes.** The unit-edge truncated cube T lies in the Voronoi cell $[-a, a]^3$ of $2a\mathbb{Z}^3$, with $a = (1+\sqrt{2})/2$. The $4 \times 4 \times 4$ grid minus its eight corners packs 56 copies in $4T$ (`tcu56`): PingYou's Cartesian shell $56 = 4^3 - 2^3$, the end of the plateau $37 \le n \le 56$.

Over the 3D catalogue, 889 records are lattice windows: every orientation class with at least four pieces sits on a lattice, with at most four classes. 685 of them have a single orientation, and 407 sit at a rational scale. Figure 3 shows how the window numbers end the plateaus. Two-orientation windows include PingYou's truncated-tetrahedron windows: $35 = 19+16$ at $11/3$ and $71 = 40+31$ at $13/3$, the two counts being coefficients of $(1+x+x^2)^4$ and $(1+x+x^2+x^3)^4$ [8].

7 Transfer by inclusion (P11)

Mechanism. Composition with a single inner piece: a packing of K' gives one of any K that fits in K', and a packing in C' gives one in any C that holds C'.
Lean. `Packing.transfer_piece`, `Packing.transfer_container`.
Prevalence. 19 Platonic records are exactly a transfer; no transfer bound beats a record.

If one copy of K fits in tK', every packing of n copies of K' in sC gives n copies of K in $(st)C$ (`Packing.transfer_piece`). If one copy of C' fits in λC, every packing of K in sC' gives one in $(\lambda s)C$ (`Packing.transfer_container`). Both are Theorem 5.1 with $m = 1$ or $n = 1$.

For the 25 Platonic families we computed the twenty one-piece scales $s_1(K \text{ in } K')$ numerically (`scripts/patterns/transfer.py`). Examples are $s_1(\text{tet in cube}) = 1/\sqrt{2}$, $s_1(\text{cube in octahedron}) = 1 + 1/\sqrt{2}$ and $s_1(\text{cube in dodecahedron}) = 1/\varphi$. We then compared every record with the best transfer bound from a neighbouring family. No transfer bound beats a record. 19 records are exactly a transfer:

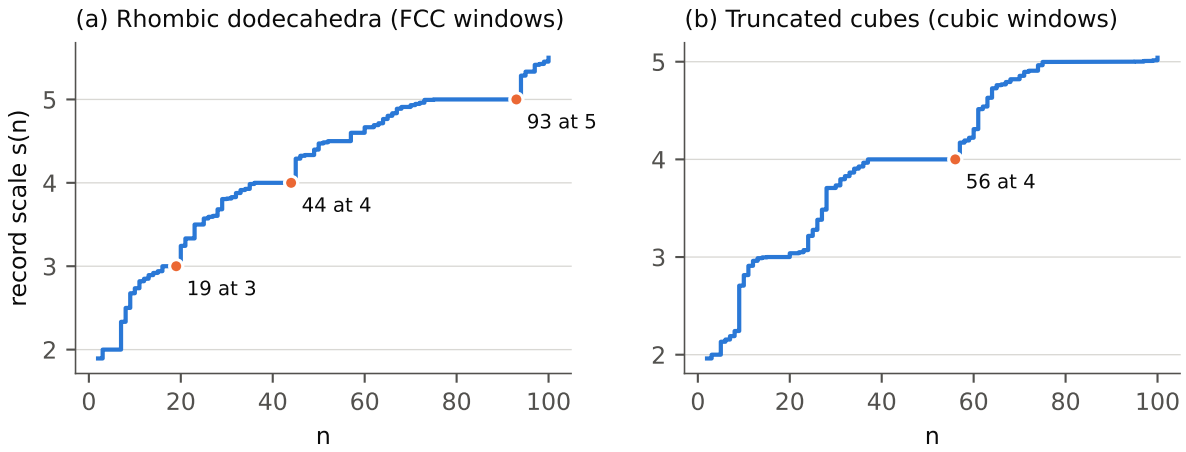


Figure 3: Record scales against n. Plateaus end at lattice window numbers: 19, 44 and 93 FCC windows of rhombic dodecahedra at scales 3, 4 and 5, and the 56-piece Cartesian shell of truncated cubes at scale 4 (all proved in Lean at the exact scales).

- cubes in a dodecahedron for $n = 6$–8, 25–27 and 61–64 are the cube grids in the dodecahedron's inscribed cube ($1/\varphi$ times the cube records);
- 2 and 3 tetrahedra in a dodecahedron and 8 cubes in an octahedron likewise sit in an inscribed cube (for two tetrahedra this is the optimal packing of Paper II [3]);
- icosahedra in a cube ($n = 8$, from $s_1(\text{ico in cube}) = \varphi$) and in an octahedron ($n = 10$ and 86–88) inherit cube and octahedron packings piece by piece.

Of the 2475 record comparisons, 25 are within 1% of the transfer bound and 181 within 5%. Transfer is therefore a consistency check on the catalogue, and a cheap source of starting points.

Part II Descriptive patterns

8 Layers and orientation classes (P6)

Pattern. Records are stacks of a few layers, with a few orientations up to the piece's symmetry, and the relative rotations between orientation classes are rational or a few algebraic angles.
Measured by. `scripts/patterns/features.py`, `rotations.py`.
Prevalence. 1,241 3D records have one orientation class, 507 have two; 3,303 of 10,444 class pairs have rational relative rotations.

Layer inventories along a container direction give the number patterns PingYou highlights. Examples are $29 = 9 + 11 + 9$ and $30 = 9 + 12 + 9$ cubes in a dodecahedron, whose shared row phase forces the scales $5/\varphi^2$ and $23 - 13\varphi$. Others are $32 = 8 + 6 \cdot 4$ cubes in an icosahedron at $8/\varphi^2$, $44 = 6 + 6 + 32$ octagonal prisms at $4 + 2\sqrt{2}$, and $20 = 2 \times 2 \times 5$ dodecagonal prisms at $15/2$ [8]. The extraction groups centres into layers along each container facet normal, and pieces into orientation classes when $R_i^{-1} R_j$ is a proper symmetry of the piece.

Relative orientations. Between two orientation classes a, b, the relative rotation $R_a^{-1} R_b$ is defined modulo the piece's proper symmetries. We take the smallest angle over that coset. Over the 1,501 3D records with two to eight classes, 10,444 class pairs give 3,303 rational angles, that is, p/q of a full turn with $q \leq 24$. In 626 records every relative rotation is rational. The rational angles are the pieces' own symmetry angles: 90°, 60°, 30°, 36°, 45°, 18°, 15° and 22.5°, in decreasing frequency. The remaining angles are not random either. They cluster at a few algebraic values, each tied to a family (Table 3). Those values compose a symmetry rotation of the piece with a symmetry rotation of the container. Examples are $\arccos(-1/4)$ and $\arccos((\sqrt{3} - 2)/4)$ for triangular prisms laid along different cube axes, and $\arccos(7/9)$ for octahedra in a cube.

angle	$\cos\theta$	class pairs	main families
93.841°	$(\sqrt{3} - 2)/4$	662	triangular prisms (heights 1, 1.5, 2, 3) in a cube
37.399°	$(4\sqrt{5} - 1)/10$	167	dodecahedra in a dodecahedron, icosahedron, octahedron
104.478°	$-1/4$	112	triangular prisms in a cube
15.631°	–	98	triangular and hexagonal prisms in a cube
95.480°	$(\sqrt{5} - 3)/8$	48	pentagonal prisms in a cube
62.799°	$(2\sqrt{2} - 1)/4$	46	cubes in a cube
38.942°	$7/9$	36	octahedra in a cube

Table 3: Recurring non-rational relative orientations between orientation classes in 3D. The cosines are matched to 3×10^{-6} and are conjectural; one cluster (a twist about the prism axis) is not identified at this precision.

9 Symmetry (P7)

Pattern. Some records are invariant under symmetries of the container.
Measured by. Exact comparison of placed vertex sets, tolerance 10^{-4} edges.
Prevalence. 776 of 8,514 3D records have a nontrivial symmetry group; the fully symmetric ones are all windows or shells.

The configuration symmetry group of a record is the set of container symmetries, reflections included, that map the set of placed pieces onto itself. Of the 8,514 3D records, 7,738 are

asymmetric and 776 have a nontrivial group (Fig. 4a). The fully symmetric records (orders 24, 48 and 120) are all windows or shells. Symmetric seeding (134 records) was one of the search moves, but most symmetric records come from lattice cuts and contributed constructions. Relaxation from a symmetric start rarely keeps the symmetry.

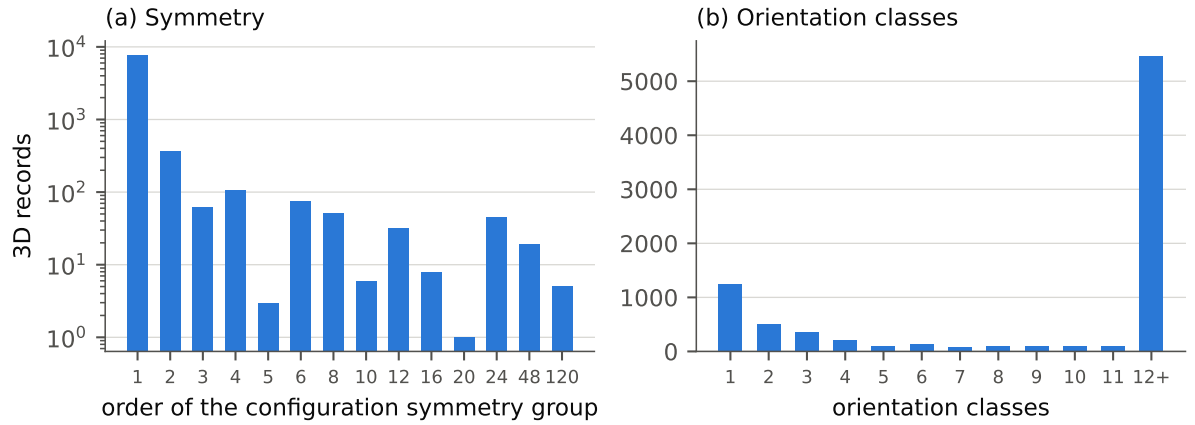


Figure 4: (a) Orders of configuration symmetry groups of the 8,514 3D records (log scale). (b) Number of orientation classes.

10 Algebraic scales (P8)

Pattern. A rigid record solves its contact system, so its scale is algebraic, in a field that reflects the contacts.
Certified by. Exact contact certificates where available; otherwise a conjectured closed form matched to 10^{-11}.
Prevalence. 1,624 closed forms: 895 rational, 266 in $\mathbb{Q}(\sqrt{2})$, 185 in $\mathbb{Q}(\sqrt{3})$, 148 in $\mathbb{Q}(\sqrt{5})$.

A rigid record is a solution of its contact system: equalities between support functions of touching pieces and walls. Its scale is then algebraic. Of the 1,624 records with a conjectured closed form, 895 are rational, 266 lie in $\mathbb{Q}(\sqrt{2})$, 185 in $\mathbb{Q}(\sqrt{3})$ and 148 in $\mathbb{Q}(\sqrt{5})$, and the rest in nested or biquadratic fields. The field reflects the geometry of the contacts. Golden scales come from dodecahedral and icosahedral walls, silver ratios $\delta = 1 + \sqrt{2}$ from square antiprisms (PingYou's $S_2 \le (5-\delta)/\sqrt{\delta}$, $S_9 \le 1+2/(11-4\delta)$, $S_{20} \le 3+1/(\delta+2)$ [9]), and $\sqrt{3}$ from hexagonal and dodecagonal prisms. A closed form is an exact statement only once the contact system is solved and certified; PingYou's "six-equation", "77-equation" and "115-equation" contact constructions are such certificates [9].

11 Contact graphs and rattlers (P10)

Pattern. The contact graph of a record, and its rattlers: pieces that touch nothing.
Measured by. `scripts/patterns/contacts.py`, contact threshold 10^{-5}.
Prevalence. 649 of 2494 3D records with $n \le 30$ (26%) contain a rattler.

The contact graph of a record joins two pieces when their best separating plane has margin below 10^{-5}, and a piece to a wall when a container facet is that close. The certified records carry 10^{-7} clearance, so touching pairs sit near 10^{-7}. A *rattler* is a piece that touches neither another piece nor a wall. Over the 2494 3D records with $n \le 30$, pieces have 3.4 piece contacts and

1.9 wall contacts on average, and 649 records (26%) contain a rattler. A rattler can be deleted without changing the scale, so its record is an $(n-1)$-configuration at scale $s(n)$.

The relations are sharp. Rattlers are rare on plateaus (lift 0.64), almost absent from lattice windows (lift 0.03, 2 records) and absent from symmetric records. They lean towards records built by insertion (lift 1.22). The 107 records whose contact count reaches the $6n+1$ degrees of freedom of the scale problem are mostly windows and plateau records (80% on a plateau, lift 3.02). Plateaus are rigid windows minus pieces, not loose packings, and a rattler marks an irregular record whose remaining pieces set the scale.

Part III Extremal patterns

12 Proven optimality (P9)

Pattern. A record whose scale is matched by a lower bound.
Lean. Papers I [2] and II [3]; the passage to plateaus is `optimal_plateau_of_common_point` and `optimal_plateau_of_one_piece`.
Prevalence. 85 records proved optimal in Lean, 27 of them for the first time; 27 more proved with computer assistance by Hosono [7].

At the extreme of the classification are records whose scale is matched by a lower bound. Papers I and II prove 85 catalogue records optimal in Lean. Four mechanisms occur, summarized in Table 4.

Lower bound	Records	Count	Paper
Volume, attained by tilings	k^d cubes ($d = 3, 4, 5$), k^3 triangular prisms	8	I, II
Cube centre lemma (Lemma F)	cubes $n = 2$–7, tesseracts $n = 2$–15, five-cubes $n = 2$–31	50	I
Facet width	two tetrahedra in four Platonic solids, two 5-simplices in a 5-cube, two 5-cells in a 24-cell	6	II
Lemma O	two octahedra in a cube	1	II
Two balls	two 16-, 24-, 120-, 600-cells in a tesseract or 16-cell	7	II
Axis widths, box centre lemma	triangular, hexagonal, octagonal and dodecagonal prisms in a cube	13	II
		85	

Table 4: The records proved optimal in Lean, by the lower bound that proves them.

Every one of them has the form of a general relation (Section 13): an extremal lower bound plus deletion proves a whole plateau.

Theorem 12.1 (`optimal_plateau_of_common_point`, `optimal_plateau_of_one_piece`)**.** *Suppose that below scale s every copy of K in the container contains a fixed point in its interior, or that below s no single copy fits, and that N copies fit at scale s. Then s is optimal for every n from 2 (respectively 1) to N.*

This explains where several plateaus start: a plateau begins where a lower bound becomes tight. The numerical survey (`scripts/optimality/`) shows that the centre argument, or the one-piece bound, is also exact for 4-, 5- and 12-gonal antiprisms and for several further prisms in a cube, and that for fifteen further Platonic pairs with $n = 2$ the centre scale equals the record to about seven digits. These are the next candidates (Paper II, §9).

Part IV Relations and dimensions

13 Relations between the classes

A record usually belongs to several classes, and the classes are not independent. Figure 5 shows both kinds of relation.

13.1 Formal implications

These are theorems in Lean, for arbitrary bodies and containers:

- a lattice window of N pieces gives every $n \leq N$ at the same scale, so windows end plateaus (`Packing.restrict`; for the four exact windows, `lattice_windows`);
- composition with one piece is transfer (`Packing.transfer_piece`, `Packing.transfer_container`);
- deletion and separator certificates make the arrows' F step sound (`Packing.restrict`, `Packing.of_separators`);
- an extremal lower bound (centre, width, box or volume lemma) together with deletion proves every record of a plateau optimal (`optimal_plateau_of_common_point`, `optimal_plateau_of_one_piece`).

13.2 Empirical associations

For classes A and B, the lift $P(B \mid A)/P(B)$ over the 3D records measures how much more often B occurs among records in A (`scripts/patterns/relations.py`). The strongest relations are:

- **The rigid cluster.** Lattice windows, a single orientation class, symmetry, composites and rational scales attract one another (lifts 3.6 to 7.2). 77% of windows have one orientation class (lift 5.29), symmetric records are windows 4.92 times as often as average, and composites have rational scales 7.25 times as often.
- **Rigidity gives plateaus.** 92% of windows and 94% of frozen subsets lie on plateaus (lifts 2.77 and 2.82), as do 82% of the records with a closed form.
- **Insertion and rattlers avoid the rigid cluster.** Records built by insertion are windows a fifth as often as average (lift 0.21), and rattlers almost never occur in windows or symmetric records.
- **Proven optimality goes with closed forms** (lift 5.41). Every proof so far is for an exact, algebraic construction.

The associations are measurements on this catalogue, not theorems. They do show where to search. Irregular records built by insertion, especially those with a rattler, are where arrows and re-optimization find gains. The rigid cluster is where exact constructions and proofs are found.

14 Rotations, symmetry and subsets in dimensions 4–6

Data. The 1,890 records of the 54 families of regular polytopes in dimensions 4–6 (`scripts/patterns/features_nd.py`, `stats_nd.py`).
Findings. Symmetry depends on the container; relative rotations in 4D are isoclinic or simple; the relations of Section 13 persist.

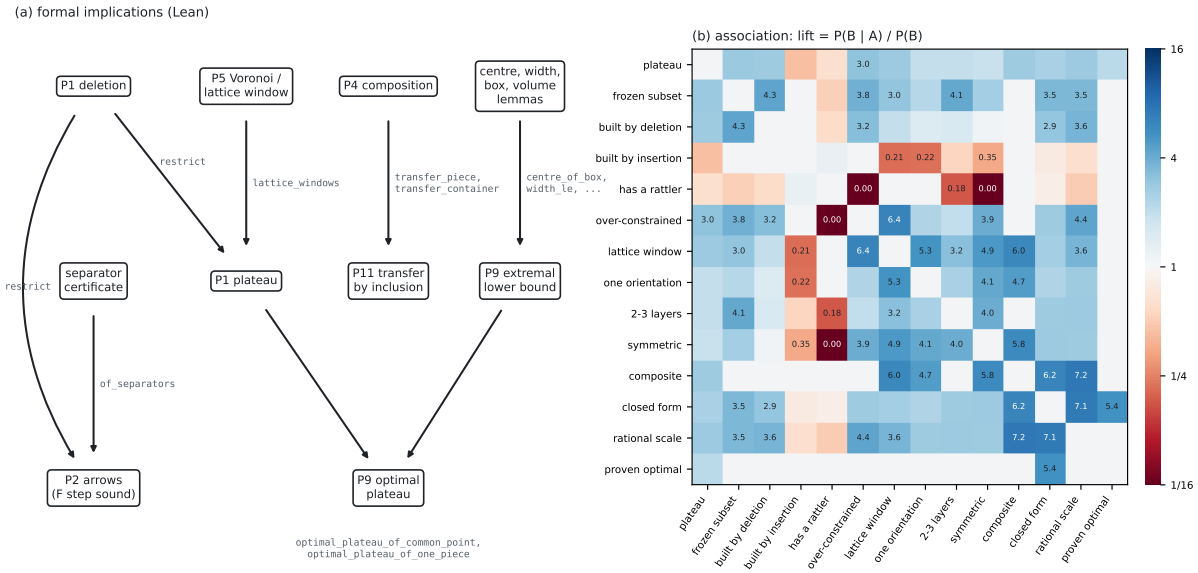


Figure 5: Relations between the pattern classes. (a) Formal implications, with the Lean theorem on each arrow. (b) Empirical lift $P(B \mid A)/P(B)$ over the 3D records (contact features for $n \leq 30$). Blue: B more frequent among records in A. Red: less frequent. White: independent. Grey: fewer than 20 joint records. Deletion and insertion, the last lineage move, are exclusive by definition.

The measures of Parts I and II apply unchanged to the higher-dimensional records. The symmetry groups are computed from the vertex sets, by backtracking over vertex images with the Gram matrix of a basis of neighbouring vertices, and checked against their orders: 120 for the 5-cell, 384 for the tesseract and 16-cell, 1152 for the 24-cell, 14 400 for the 600- and 120-cells, 720, 3840 and 46 080 for the simplex, cube and cross-polytope in 5D and 6D, and 5040 for the 6-simplex. A relative rotation $R_a^{-1}R_b$ is reduced modulo the piece's proper symmetries to the representative closest to the identity, and described by its rotation angles, one per invariant plane. It is *simple* if one angle is nonzero and *isoclinic* if all are equal.

	3D	4D	5D	6D
records	8,514	1,404	279	207
one orientation class (%)	14.6	17.5	15.4	15.9
nontrivial configuration symmetry (%)	9.1	20.0	22.6	24.6
on a plateau (%)		21.1	17.6	22.7
consecutive records frozen (%)	17.9	16.4	16.7	19.7
rational relative rotations (%)		12.6	1.1	1.4
lattice windows	889	264	41	12

Table 5: The pattern measures by dimension. The 3D plateau and rotation figures use other denominators (Sections 2 and 8) and are left out.

Table 5 compares the dimensions. Three observations stand out.

14.1 Symmetry depends on the container

Records in a simplex are almost never symmetric (5-cell 6.8%, 5-simplex 5.4%, 6-simplex 8.7%), and few have a single orientation class. Records in a cube are the most symmetric (tesseract 32.9%, 5-cube 36.6%, 6-cube 36.2%), more than three times the 3D rate (Fig. 6). Some records carry the container's whole group: 25 24-cells in a 24-cell (order 1152), and 10 and 12 cross-

polytopes in the 5- and 6-cross-polytope (orders 3840 and 46 080), with one piece towards each vertex of the container and, in the 24-cell, one more at the centre.

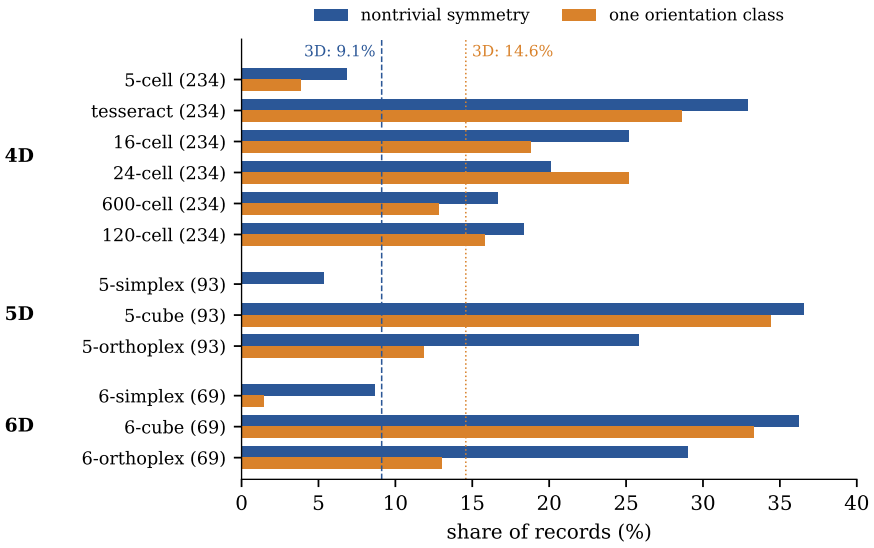


Figure 6: Records in dimensions 4–6 with a nontrivial configuration symmetry (blue) and with a single orientation class (orange), by container; the number of records is in parentheses. Dashed and dotted lines: the 3D rates. Simplex containers are almost never symmetric; cube containers are the most symmetric in every dimension.

14.2 Relative rotations in 4D are isoclinic or simple

The commonest rational relative rotations are the isoclinic $(60°, 60°)$ (96 class pairs), the simple 45° (70), the simple 72° (48), the isoclinic $(45°, 45°)$ (35) and the simple 90° (29). Figure 7 shows all class pairs. The 60° isoclinic rotations are left multiplications by unit quaternions of order 3 or 6, which permute the three 16-cells inscribed in a 24-cell. Accordingly, 24 16-cells in a 24-cell (scale $\sqrt{2}$, symmetry 1152) and 24 16-cells in a 600-cell come in three classes of eight. The 72° rotations come from the 600- and 120-cells. In 5D and 6D almost no relative rotation is rational.

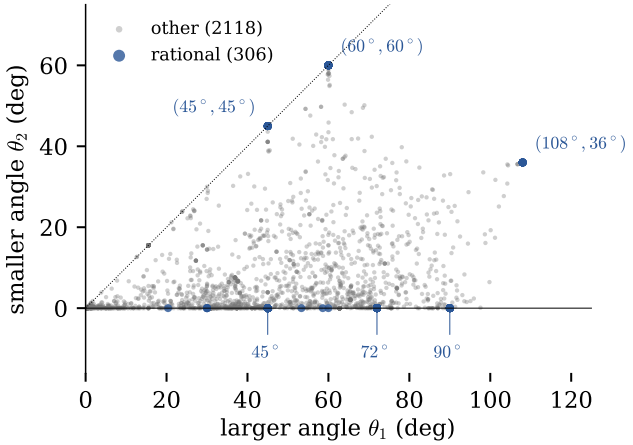


Figure 7: The two rotation angles $\theta_1 \geq \theta_2$ of the relative rotation between each pair of orientation classes, over the 4D records. Simple rotations lie on the axis $\theta_2 = 0$, isoclinic ones on the diagonal. Blue: rational angles, with the commonest labelled.

14.3 The relations persist

Over the 4D–6D records the lifts are 3.8 (symmetric given one orientation class), 2.8 (lattice given symmetric), 3.7 (plateau given lattice), 3.2 (plateau given symmetric) and 3.4 (closed form given symmetric). About one consecutive pair in six is a frozen subset in every dimension. The most frozen pairs are in the cube families and the 24-cell, whose records are grids and shells.

15 PingYou's sixteen constructions in the classification

Table 6 places the sixteen constructions of PingYou's notes [8] in the classification. All are in the catalogue. Four have since been improved by `konoha-ily2`, so they are historical in the sense of their notes. Six are proved in Lean: five at their exact scales, and the 64 triangular prisms as optimal. The others are certified records.

pieces, container	count	exact scale	class	catalogue now	formal status
truncated cubes	$61 = (4^3 - 8) + 5$	$13 - 6\sqrt{2}$	P3 completion of P5	PingYou Ltd	certified
cubes, icosahedron	$32 = 8 + 6 \cdot 4$	$12 - 4\sqrt{5}$	P6 shell	konoha-ily2 (3.0291)	certified
cubes, dodecahedron	$40 = 14 + 14 + 6 + 6$	$5\varphi - 6$	P6 layers	PingYou Ltd	certified
cubes, dodecahedron	$29 = 9 + 11 + 9$	$5/\varphi^2$	P6, P8	konoha-ily2 (1.8595)	certified
cubes, dodecahedron	$30 = 9 + 12 + 9$	$23 - 13\varphi$	P6, P8	konoha-ily2 (1.9022)	certified
triakis tetrahedra	$16 = 4^2$	$(5/3)^2$	P4 recursion	PingYou Ltd	Lean (exact)
triakis tetrahedra	$64 = 4^3$	$(5/3)^3$	P4 recursion	PingYou Ltd	Lean (exact)
triangular prisms	$64 = 4^3$	4	P5 tiling, P9	PingYou Ltd	Lean (optimal)
truncated tetrahedra	$71 = 40 + 31$	13/3	P5 two orientations	PingYou Ltd	certified
truncated tetrahedra	$35 = 19 + 16$	11/3	P5 two orientations	PingYou Ltd	certified
truncated cubes	$56 = 4^3 - 2^3$	4	P5 Cartesian shell	PingYou Ltd	Lean (exact)
rhombic dodecahedra	$19 = 1 + 12 + 6$	3	P5 FCC shell	PingYou Ltd	Lean (exact)
rhombic dodecahedra	$44 = 2 + 8 + 12 + 12 + 8 + 2$	4	P5 FCC window	PingYou Ltd	Lean (exact)
height-2 triangular prisms, cube	$21 = 6 + 15$	3	P6 mixed layers	PingYou Ltd	certified
octagonal prisms, cube	$44 = 6 + 6 + 32$	$4 + 2\sqrt{2}$	P6, P8	PingYou Ltd	certified
height-3/2 dodecagonal prisms, cube	$20 = 2 \cdot 2 \cdot 5$	15/2	P5 column array	PingYou Ltd	certified

Table 6: PingYou's sixteen constructions [8]: number pattern, exact scale, class, current catalogue holder (with the current scale where it changed), and formal status.

16 Classification by family

Table 7 classifies the records by family group. Prisms and antiprisms, and the Platonic pairs, are dominated by plateaus and lattice windows: their pieces tile or nearly tile. The Catalan and Archimedean families are mostly irregular, with fewer windows and fewer frozen subsets. Their records are individually tightened contact configurations, where arrows are the natural tool. In dimensions 4–6, plateaus and closed forms are common, coming from grids, composites and the $n = 2$ centre lemmas. Figure 8 shows how the records were produced, and Fig. 9 shows examples.

Group	families	records	on a plateau	closed form	symmetric	lattice window	one orientation	frozen subsets
Platonic pairs	25	2,475	791	272	274	347	441	327
Archimedean	13	1,287	227	120	108	124	212	132
Catalan	13	1,287	99	72	110	42	215	50
Prisms and antiprisms	35	3,465	1,892	742	284	376	373	875
4D regular polytopes	36	1,404	297	300	–	–	–	–
5D regular polytopes	9	279	49	64	–	–	–	–
6D regular polytopes	9	207	47	54	–	–	–	–

Table 7: Pattern prevalence by family group: records on a plateau, with a closed form, with nontrivial symmetry, that are lattice windows, with a single orientation class, and frozen-subset pairs $f(k, N) = 1$ $(N - k \leq 6)$. Geometric features are computed for 3D only.

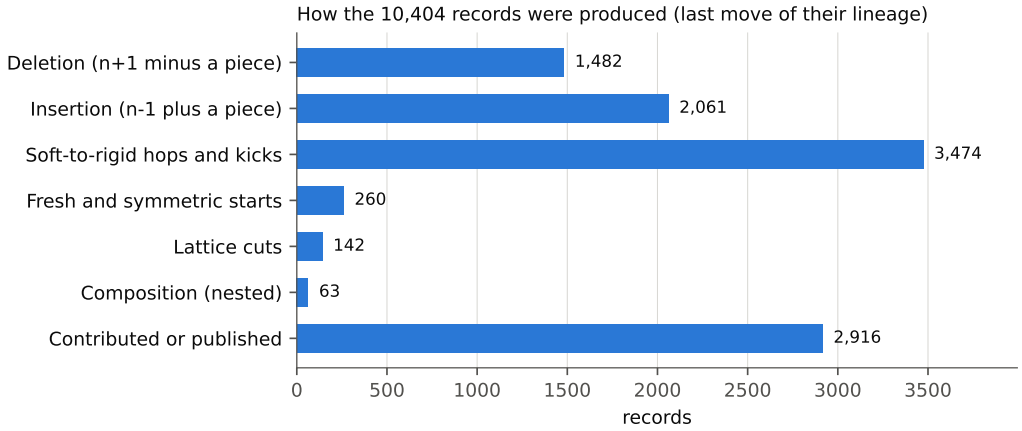


Figure 8: The last move in each record's lineage. Deletion and insertion together produced a third of the catalogue.

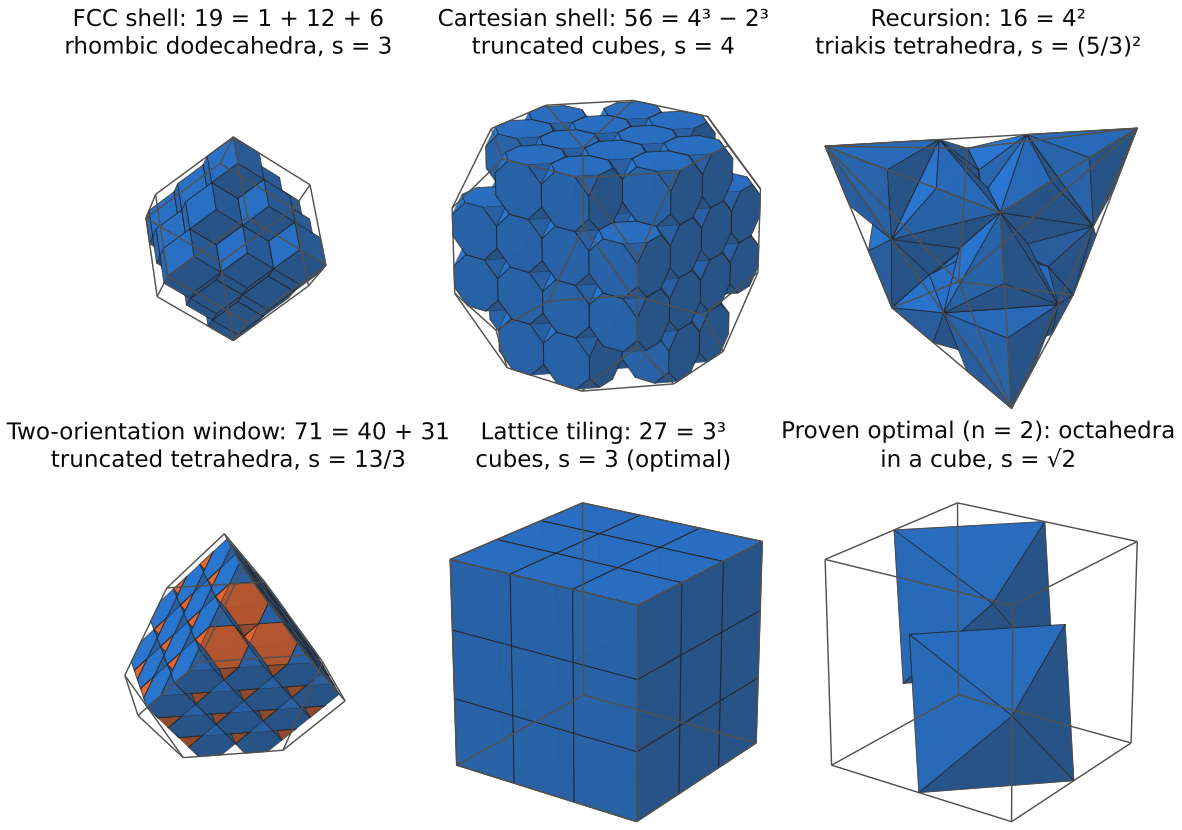


Figure 9: Pattern classes in records. Top: an FCC shell, a Cartesian shell and the triakis recursion (all proved in Lean at the exact scale). Bottom: a two-orientation window (orientation classes in blue and orange), a cube tiling, and a proven-optimal $n = 2$ record.

Part V Formalization and limits

17 Formalization

All theorems cited above are in `lean-optimality/` (Lean 4.31, Mathlib). The continuous-integration audit checks that every main theorem depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`. Table 8 lists the files of this paper; the optimality files are described in Papers I and II.

File	Main declarations	Content
`Basic.lean`	`Packing`, `Packing.restrict`	packings, deletion
`Patterns.lean`	`Packing.mono`, `Packing.compose`, `Packing.selfCompose`, `Packing.pow`	scale monotonicity, composition
	`Packing.of_separators`, `Packing.smulBoth`	separator certificates (F), rescaling
	`voronoi_disjoint`, `voronoi_packing`, `lattice_window`	Voronoi windows
`Windows.lean`	`rdo_window`, `tcu_window`, `rdo19`, `rdo44`, `rdo93`, `tcu56`	exact FCC and cubic windows
`Triakis.lean`	`ktt4`, `triakis_recursion`, `ktt16`, `ktt64`	the recursion 4^k at $(5/3)^k$
`Relations.lean`	`Packing.transfer_piece`, `Packing.transfer_container`, `optimal_plateau_of_common_point`, `optimal_plateau_of_one_piece`	transfer; extremal bound plus deletion

Table 8: The Lean files of this paper, in `lean-optimality/PackingOptimality/`.

The formal results are of three kinds. The mechanisms and the relations between them are proved for arbitrary bodies and containers: deletion, monotonicity, composition, transfer, separator certificates, Voronoi windows and the passage from an extremal bound to an optimal plateau. The exact instances are specific packings at exact algebraic scales; the catalogue certifies their padded counterparts by exact arithmetic. The optimality results are lower bounds matched by exact constructions, in Papers I and II. Descriptive classes (layers, orientation classes, symmetry, contacts) are measurements, not theorems.

18 Limitations

The classification is empirical: a record can belong to several classes, and the thresholds (tolerances, the cap of four orientation classes for windows, the gap $N - k \leq 6$ for inheritance) are choices. On square antiprisms the arrows reproduce values PingYou Ltd announced first and produced no bound below theirs; their submitted witnesses are now the records. Contact graphs depend on the 10^{-5} threshold. The one-piece scales behind the transfer bounds are numerical, so the transfer comparison is a measurement, although each transfer construction, once made explicit, is certified like any other record. Closed forms are conjectures until their contact systems are solved exactly. The formal layer proves the mechanisms and the listed instances, not the optimality of the windows. Only the 85 records of Section 12 are proven optimal.

Acknowledgements

The arrows of Section 3, their F and P steps, and many of the number patterns of Sections 5, 6 and 8 are due to PingYou Ltd [8, 9]. Their constructions also include the corner completion, golden shells and layers, finite Welsh windows, FCC shells and windows, the Cartesian shell, and mixed, quadratic-field and column layers. The catalogue combines searches and contributions

by PingYou Ltd, `konoha-ily2` (Hirotaka Nakano), PolyPack (Michael Sinko), Berthold, Kamp, Mexi, Pokutta and Pólik [6], Hiroaki Hosono [7], Stian, Ryan Xu and others. Soft-to-rigid continuation is due to Yohei Nakajima [5]. AI coding and drafting tools, including Claude (Anthropic), assisted implementation, analysis, formalization and manuscript preparation under the author's direction. The Lean build and the exact certificates, not the tools, are the evidence for the formal and certified claims.

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